

End-to-end modeling of kilonova light curves and gravitational waveforms from neutron star mergers for possible constraints on nuclear physics parameters

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics
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ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. Although measurements of neutron-star masses, radii, and tidal deformabilities have constrained the pressure of dense matter, there is still no consensus on the composition of the stellar core, in particular on whether a phase transition to deconfined quark matter takes place. Binary neutron-star mergers (BNSMs) probe even higher densities and temperatures and, as demonstrated by the gravitational-wave event GW170817 and its associated kilonova, are a site of rapid neutron-capture (r-process) nucleosynthesis. In these events, the EOS shapes the hydrodynamic evolution of the merger and may therefore leave fingerprints both in the gravitational waveform and in the mass and initial thermodynamic conditions (e.g., density and entropy) of the ejected material. The subsequent nucleosynthesis in the ejecta, however, also depends on the properties of neutron-rich nuclei far from stability, including nuclear masses, β^- -decay and neutron-capture rates, and fission properties. Because dense-matter and nuclear-physics uncertainties affect the same observables, disentangling them is essential to determine what current and future multimessenger observations can reveal about dense matter. In this work, we perform a joint analysis of the gravitational-wave and electromagnetic signatures of BNSMs to investigate whether a hadron-quark phase transition leaves an imprint on the gravitational-wave signal and on the ejected material, whose mass and composition determine the radioactive heating that powers the kilonova. To this end, we perform general-relativistic hydrodynamic simulations of an equal-mass BNSM with the Einstein Toolkit, using a hadronic EOS, a hybrid EOS that coincides with it below the onset of a first-order hadron-quark phase transition, and an alternative hadronic EOS. Since the inspiral is identical for the hadronic and hybrid models by construction, post-merger differences can be attributed to the phase transition, while the alternative hadronic EOS allows us to assess whether the combined gravitational-wave and electromagnetic signatures can distinguish a phase transition from a change in the stiffness of hadronic matter. We compare the post-merger gravitational-wave signals and the dynamical ejecta, whose thermodynamic evolution we post-process with the nuclear reaction network WinNet to obtain r-process abundances and radioactive heating rates, from which we compute kilonova light curves.

Finally, we quantify the impact of the phase transition relative to the variation of the hadronic EOS, to uncertainties in nuclear-physics inputs such as fission-fragment distributions and β^- -decay rates, and to numerical resolution.

Thesis supervisor: Neelima Govind Kelkar

Title: Professor of Physics

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Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.

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Chapter 1

Introduction

Here we will responds:

1. What question I am trying to answer?
Can the joint analysis of GW and EM signals encode relevant information about the existence of the phase transition inside a neutron star?
2. Why is this question open an relevant?
The possible confirmation or dismiss of quark-gluon plasma inside neutron stars could help us stablish the structure of neutron stars and test theoretical results in that regime.
3. What will I do to contribute to this question?
This thesis is just a prove of concept and a first approach to use the two different signals from a merger to try to identify possible encode information about the nature of the matter in the merger.

1.1 Dense matter and neutron-star equation of state

This section concentrates on the context of the second question. Here we will have a overview of the known equations of state, what they account for and their limitations.

1.2 Binary neutron-star mergers as multimessenger laboratories

This section is the state of the art of current efforts to use compact object mergers, in particular neutron star mergers, as test for theoretical develoment.

1.3 Open problems: phase transitions and nuclear-physics uncertainties

This section as continuation of the first one states why is so difficult to calculate the equation of state of matter inside a neutron star.

1.4 Objectives and research questions

This section focus on the first question, it states what question am i answering. Are combine GW and EM be usefull for testing theretical results?

1.5 Structure of the thesis

This is and overall review of what will i do in this thesis (answer to question 3)

Chapter 2

Dense Matter and Neutron-Star Structure

What we know about the state of the matter inside a neutron star. What theoretical result we know almost for sure about the structure and what implies the equations to the star deformability, radii and mass.

2.1 Equation of state of cold, β -equilibrated matter

An overview of the matter conditions inside of the neutron star.

2.2 Hadron–quark phase transitions and hybrid equations of state

What a phase transition should be like and why could this happen?

2.3 Thermal effects in the equation of state

If a phase transition is happening what does that imply to the neutron star?

2.4 Stellar structure: TOV equations and tidal deformability

Given the equations for the curvature of a neutron star and a certain equation of state what should we expect about tidal deformability?

2.5 Observational constraints

Given all the observations done What have we learned about the actual nature of those stars?

Chapter 3

Binary Neutron-Star Mergers

- 3.1 Merger phases and remnant fate
- 3.2 Gravitational waves: inspiral tidal effects and post-merger spectrum
- 3.3 Mass ejection: tidal, shock-driven, and disk-wind ejecta
- 3.4 Weak interactions and the electron fraction of the ejecta
- 3.5 Electromagnetic counterparts

Chapter 4

Numerical Relativistic Hydrodynamics

- 4.1 3+1 decomposition and evolution formulation
- 4.2 General-relativistic hydrodynamics in conservative form
- 4.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere
- 4.4 Gravitational-wave extraction
- 4.5 Ejecta diagnostics and unbound-matter criteria

Chapter 5

r-Process Nucleosynthesis

- 5.1 Nuclear statistical equilibrium and freeze-out
- 5.2 The r-process: neutron-to-seed ratio, fission cycling
- 5.3 Reaction-network equations and numerical solution
- 5.4 Nuclear inputs: masses, decay rates, neutron capture, fission
- 5.5 Radioactive heating and thermalization

Chapter 6

Kilonova Light Curves

6.1 Radiative transfer and the diffusion regime

6.2 Opacities of r-process matter

6.3 Light-curve modeling

Chapter 7

Methods

- 7.1 Hadronic and hybrid equation-of-state models
- 7.2 Binary neutron-star simulation setup
- 7.3 From simulation to trajectories: ejecta extraction and expansion
- 7.4 Network setup and nuclear-physics variations
- 7.5 Kilonova model setup
- 7.6 Gravitational-wave analysis and detectability metrics

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.

Chapter 8

Results

8.1 Stellar properties of the equation-of-state models

8.2 Numerical accuracy and convergence

8.3 Merger dynamics and remnant evolution

8.4 Gravitational-wave signals

8.5 Dynamical-ejecta properties

8.6 Nucleosynthesis and heating rates

8.7 Kilonova light curves

8.8 Joint GW EM signatures

8.9 Uncertainty budget: EOS, nuclear physics, and resolution

8.10 Detectability with current and future facilities

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Chapter 9

Discussion

9.1 Comparison with previous studies

9.2 Limitations and model assumptions

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Chapter 10

Conclusions and Outlook

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Appendix A

Units and conventions

Appendix B

Convergence tests

Appendix C

Network and equation-of-state configuration details